Frequency modulates structural choice in Turkish suspended affixation: a latent-process account

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Abstract

Suspended affixation (SA) allows a suffix on one conjunct to scope over all coordinated elements. While inflectional SA is productive in Turkish, derivational SA is claimed to be highly restricted; yet speakers readily accept certain cases. We propose that this gradient acceptability reflects a frequency-modulated choice between two possible syntactic representations: base-generation, which licenses derivational SA, and ellipsis. To test this, we conducted a rating task on the acceptability of four derivational suffixes in SA form while manipulating the frequency of coordinations. Using a Multinomial Processing Tree model to isolate latent structural choices from surface ratings, we found that frequency modulated SA acceptability for some suffixes (i.e., *-sIz* '-less' and *-cI* '-maker'), but not others (i.e., *-lI* '-having' and *-lIk* '-for'). These findings suggest that frequency shapes syntactic parsing in morphologically complex environments.

1 Introduction

A central puzzle in the morphosyntax of agglutinative languages concerns the distribution of affixes across coordinated structures. In Turkish, a single suffix appearing on the rightmost conjunct can be semantically interpreted as scoping over multiple conjuncts—a phenomenon termed *suspended affixation* (SA). Consider (1), where the dative marker *-ye* surfaces only on the second conjunct yet receives an interpretation in which both *Ali* and *Ayşe* bear dative case.

(1) Ali ve Ayşe-ye bak-tı-m.
Ali and Ayşe-DAT look-PST-1SG
'I looked at Ali and Ayşe.'

This construction has been documented across typologically diverse languages, including Turkish (Kornfilt, 1996, 2012; Kabak, 2007; Broadwell, 2008; Akkuş, 2016; Atmaca, 2021), Mari (Guseva and Weisser, 2018), Ossetic, Iron, and Eastern Armenian (Erschler, 2012), Dagur (Gong, 2021), Japanese and Korean (Yoon and Lee, 2005), Nivkh (Gruzdeva, 1998), and Hungarian (Trommer, 2008).

While SA with *inflectional* morphology is relatively uncontroversial, the availability of *derivational* SA remains theoretically contested. While some suggest that derivational SA is not a productive process in Turkish (Atmaca, 2021; Kabak, 2007; Kornfilt, 2012), others suggest that it is more productive than previously assumed (Akkuş, 2016). Drawing on these conflicting observations, the current study empirically examines whether the gradient acceptability of derivational SA in Turkish is modulated by the frequency of coordinations. By fitting a Multinomial Processing Tree model to rating data on SA forms across different suffixes and coordinations with varying frequencies, we show evidence that frequency does not simply increase surface acceptability of all suffixes, but instead influences syntactic parsing for certain suffixes.

1.1 The status of derivational SA in Turkish

Earlier observations regarding derivational SA in Turkish report that it is a rare phenomenon with considerable inter-speaker and inter-item variation. For example, Kabak (2007) argues that apparent cases of derivational SA, such as (2), are restricted to tightly bound, lexicalized collocations (see also Atmaca, 2020).

(2) ay-yıldız-lı bayrak
moon-star-ADJ flag
'a flag with a moon and star' (lit. 'a moon-star-ADJ flag')

Under his view, examples such as (2) are idiomatic chunks stored holistically in the lexicon and denoting frequently co-occurring entities (Wälchli, 2005), rather than reflecting a productive affixation process. Similarly, Kornfilt (2012) has argued

that SA in Turkish is strictly limited to inflectional affixes.

However, subsequent work by Akkuş (2016) challenges this categorical restriction. Based on examples sourced online, Akkuş demonstrates that derivational SA across both nominal and verbal domains is more productive than previously thought. Crucially, he shows that the distributing affix can scope over full phrasal conjuncts—not merely adjacent, tightly-bound lexical items—e.g., '[5 lira] and [10 dollar]-DER' in (3).

(3) ... beş lira ve on dolar-lık
five lira and ten dollar-DER
banknot-lar...
banknote-PL

'5-lira and 10-dollar banknotes'

Moreover, some cases of derivational SA cited by Akkuş are acceptable regardless of the order of conjuncts, shown for the conjunct *dost ve arkadaş* 'buddy and friend' along with its version with a reversed order in (4). This flexibility is hard to reconcile with the idea that these SA constructions are simple isolated collocations stored in the lexicon.

(4) a. Bütün eğitim çalışmaları boyunca
all training sessions during
dost ve arkadaş-ça bir ortam
buddy and friend-DER an atmosphere
ol-malı-dır.
be-must-MOD

'There must be a friendly and intimate environment during the whole training session.'

b. ... buna uygun davran-mak
this-DAT appropriate behave-NMLZ
için arkadaş ve dost-ça
for friend and buddy-DER
gel-di-k.
come-1PL-PST

'... accordingly, we came in a friendly and intimate manner.'

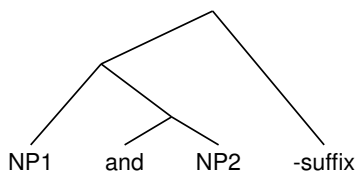
At the same time, the fact remains that derivational SA is more idiosyncratically restricted than inflectional SA. Specifically, we observe that it appears to be more readily accepted for high-frequency coordinations such as *moon and star* and *salt and lemon*, as correctly noted by Kornfilt and Kabak.

While Akkuş's findings have been adopted in subsequent work on SA (e.g., Atmaca, 2021; Türk, 2025), the role of frequency in modulating the availability and parse of these constructions has not been systematically investigated. Given the speaker- and item-level variation noted in the literature, we propose that the apparent conflict between the intuitions of Kabak (2007) and the evidence provided by Akkuş (2016) may be reconciled by attending to the *probabilistic* nature of syntactic choice underlying derivational SA.

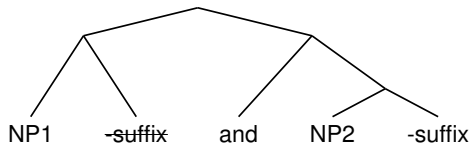
1.2 The present study

Our proposal in this study rests on the assumption that in general, SA is structurally ambiguous between two possible parses. Under a *base generation* analysis as in (5), the suspended affix combines with the coordinated noun phrase as a whole. Under an *ellipsis* analysis as in (6), the affix is underlyingly merged with each conjunct, but is deleted in the non-final conjunct(s). When encountering an SA construction, speakers are in principle free to adopt either parse.

(5)



(6)



Against this background, we argue that the difference in productivity between derivational and inflectional SA can be accounted for as follows. First, we propose that derivational affixes can only participate in SA in a base generation structure and not in an ellipsis structure, whereas inflectional affixes can freely undergo ellipsis. What distinguishes derivational from inflectional affixes is that they sit closer to the root and are more tightly integrated with it semantically, phonologically and prosodically. We suggest that as a result, derivational affixes cannot be deleted at the exclusion of the root since the two form too inseparable a unit.

Second, we argue that the availability of the two structural paths to SA in (5) and (6) is shaped by the frequency of the coordinated structure. There is strong evidence that speakers store holistic representations of coordinated noun phrases, and that the more frequent the coordination is, the more likely speakers are to access this holistic representation in parsing (Morgan and Levy, 2015, 2016). Crucially, such a holistic representation is only compatible

with a base generation analysis; in the ellipsis structure, the coordinated phrase such as *mother and father* does not form a constituent. Consequently, we propose that high-frequency structures bias the parser towards a base generation analysis.

Taken together, these two proposals account for the differences in productivity of inflectional and derivational SA. Since derivational SA is only available with a base generation analysis, it is more acceptable with high-frequency coordinations, as has been observed anecdotally in the literature and as we confirm experimentally in the following. Upon encountering a high-frequency coordinated phrase, speakers tend to access the holistic representation associated with it. This leads them to posit a base generation analysis, which is compatible with derivational SA. In contrast, inflectional SA is highly productive overall (Atmaca 2020, but see Kornfilt 1996, Türk 2025). This is as expected if inflectional affixes can freely participate in SA under both structural analyses. Even if speakers posit an ellipsis analysis of SA for low-frequency coordinations, the result is still acceptable.

Our claim that the frequency of coordinated noun phrases modulates structural choice in SA is rooted in much previous work showing evidence that frequency distributions can influence online parsing. For example, a large body of evidence supports the claim that frequency affects the processing of both single words and complex multi-word expressions (Pinker and Ullman, 2002; Arnon and Snider, 2010; Carroll and Conklin, 2020; Siyanova-Chanturia et al., 2011; Sosa and MacFarlane, 2002; Swinney and Cutler, 1979; Tremblay et al., 2011; Morgan and Levy, 2015, 2016). Frequent multi-word sequences are recognized faster and processed more fluently than matched infrequent sequences, even when individual word frequency is controlled. This suggests that speakers maintain holistic representations for recurrent phrases alongside compositional parsing.

Morgan and Levy (2015, 2016) show evidence that frequency-based storage interacts with other grammatical preferences. They show that more frequent coordinated phrases exhibit more rigid ordering preferences. For example, *ladies and gentlemen* occurs almost exclusively in this order, whereas *towels and sunglasses* alternates freely with *sunglasses and towels*. Morgan and Levy interpret this as a trade-off between compositional and holistic representations: as a coordination becomes more frequent, speakers increasingly store it as a unit rather than assembling it compositionally, resulting in frozen ordering.

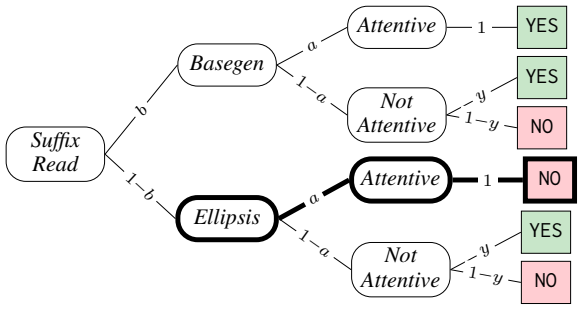
A key finding that is relevant for the present study is that the effect of frequency does not have to be linear. Alternatively, the effect of frequency can be seen as a choice between competing syntactic structures, as has been suggested for the domain of argument structure alternations, most notably the English dative alternation. Bresnan and colleagues (Bresnan, 2007; Bresnan and Ford, 2010; Bresnan et al., 2004) have shown that the choice between a double object construction (*I gave Mary the book*) and a prepositional dative (*I gave the book to Mary*) is probabilistic rather than categorical, and is governed by a range of factors including animacy, givenness, and constituents' phonological weight. Crucially, verb bias—the frequency with which a verb appears in one construction over the other—shapes expectations about structural form and influences both production and comprehension, with speakers' naturalness ratings closely tracking corpus-derived construction probabilities (Bresnan, 2007).

Similarly, our proposal goes beyond the claim that frequent expressions are simply stored as unanalyzed wholes. While we draw on Morgan and Levy's idea that higher frequency coordinations are more likely to be represented holistically, our critical prediction concerns not storage *per se*, but its downstream consequences for syntactic structure assignment. In our model, the key variable modulated by frequency is the probability of adopting a base-generation parse over an ellipsis parse. Highly frequent coordinations are more likely to be accessed holistically, making the base-generation structure—in which the affix attaches to the coordination as a single constituent—more readily available. Lower-frequency coordinations remain compatible with both routes, but the ellipsis parse—in which the affix merges with each conjunct separately—becomes relatively more available. Thus, frequency does not directly boost acceptability; instead, it shifts the distribution over latent structural analyses, and these structural analyses in turn determine whether a given affix type can participate in SA.

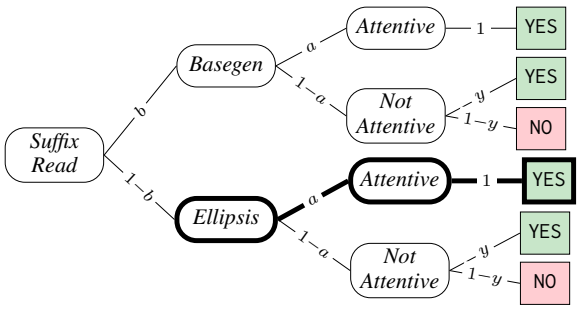
More specifically, in the current study, we present a formal hypothesis that captures how frequency and structural ambiguity interact to yield the gradient acceptability profile observed in Turkish derivational SA. We formalize this via Multinomial Processing Trees (Riefer and Batchelder,

1988; Erdfelder et al., 2009), in which we model the observed acceptability responses as a process generated by a sequence of latent subprocesses rather than by a single undifferentiated acceptance probability (Riefer and Batchelder, 1988; Batchelder and Riefer, 1999; Klauer, 2010). For the case we present here, the relevant subprocesses are structural choice (base generation vs. ellipsis), attentional state (i.e., when participants are not attending to the stimulus), and a residual yes-response tendency in an inattentive state. The trees in (7) and (8) illustrate the logic schematically. In the derivational case, attentive processing leads to acceptance under the base-generated parse but not under the ellipsis parse; in the inflectional case, both structural routes remain compatible with acceptance. In both trees, inattentive trials fall back to a participant-specific yes-bias. We predict that if frequency modulates structural choice in parsing derivational SA, we should observe a higher probability for a base generation route with more frequent coordinations.

(7) Derivational Suffix



(8) Inflectional Suffix



Next, Section 2 presents experimental data confirming a frequency effect for SA with certain kinds of derivational affixes. Section 3 then introduces a Multinomial Processing Tree model of the results.

2 Experiment

We conducted a Likert-scale judgment task to test whether the acceptability of SA involving derivational suffixes is modulated by the frequency of the coordinated phrases. We predicted that if derivational SA is sensitive to frequency, the acceptability of coordinated phrases with SA would increase as a function of their frequency. Furthermore, if this sensitivity is a general property, we should observe comparable frequency effects across all tested suffixes.

2.1 Materials and methods

The stimuli were Turkish sentences including a coordinated phrase as a modifier of a noun phrase in the sentence. We employed a 2x4 within-subject design, crossing SA, i.e., coordinated phrases with SA (i.e., with the derivational suffix appearing only on the second conjunct) or in FULL form (i.e, with the derivational suffix appearing on both conjuncts), and SUFFIX TYPE, i.e., *-cI* '*-maker*', *-lI* '*-having*', *-lIk* '*-for*' or *-sIz* '*-less*'. All four suffixes are highly productive in forming modifiers in the language. Example stimuli are in Table 1.

There were 32 experimental sentences, each with a different coordinated noun phrase. The coordinated phrases covered a wide range of frequency: their counts were taken from a Turkish Web 2012 corpus (∼3.4 billion words), and ranged from 131,657 ('mother and father') to 1 ('mouse and lion'). To ensure semantic naturalness, coordinated nouns were taken from the same conceptual realm.

The participants of the task were 47 Turkish speakers. Their task was to judge the acceptability of sentences on a 6-point Likert scale (6='very good'). The experiment had 32 experimental sentences, interspersed with 64 filler items. The task was conducted online on PCIbexFarm (Zehr and Schwarz, 2018).

2.2 Results

Figure 1 shows the observed mean ratings as a function of centered rank frequency, with higher values corresponding to more frequent coordinations. Descriptively, neither the SA penalty nor the effect of frequency seems to be uniform across suffixes.

To analyze the ordinal rating data, we fit separate Bayesian cumulative-logit models for each suffix using `brms` (Bürkner, 2017). The models included fixed effects for SA, centered rank frequency, and their interaction, as well as varying effects for subjects and items. SA was sum contrast coded, with .5 for FULL and −.5 for SA forms. Figure 2 plots the posterior means, their 89% intervals and the posterior probability of a positive effect for SA and its

SUFFIX	Sentence								
-cI '-maker'	Selin	tamir	için	kapı(-cı)	ve	pencere-ci	bir	marangozla	anlaşmış.
	S.	repair	for	door-MAKER	and	window-MAKER	a	carpenter	made.a.deal.with
	'Selin agreed with a door-maker and window-maker carpenter for the repair.'								
-lI '-having'	Selin	tamir	için	kapı(-lı)	ve	pencere-li	bir	atölye	hazırlamış.
	S.	repair	for	door-HAVING	and	window-HAVING	an	atelier	prepared
	'Selin prepared an atelier with a door and a window for the repair.'								
-lIk '-for'	Selin	tamir	için	kapı(-lık)	ve	pencere-lik	bir	desen	seçmiş.
	S.	repair	for	door-FOR	and	window-FOR	a	pattern	chose
	Selin chose a pattern for a door and a window for the repair.								
-sIz '-less'	Selin	tamir	için	kapı(-sız)	ve	pencere-siz	bir	atölye	hazırlamış.
	S.	repair	for	door-LESS	and	window-LESS	an	atelier	prepared
	'Selin prepared an atelier without a door or a window for the repair.'								

Table 1: Example stimuli from the acceptability rating task. Suffixes in parantheses are omitted in forms with SA.

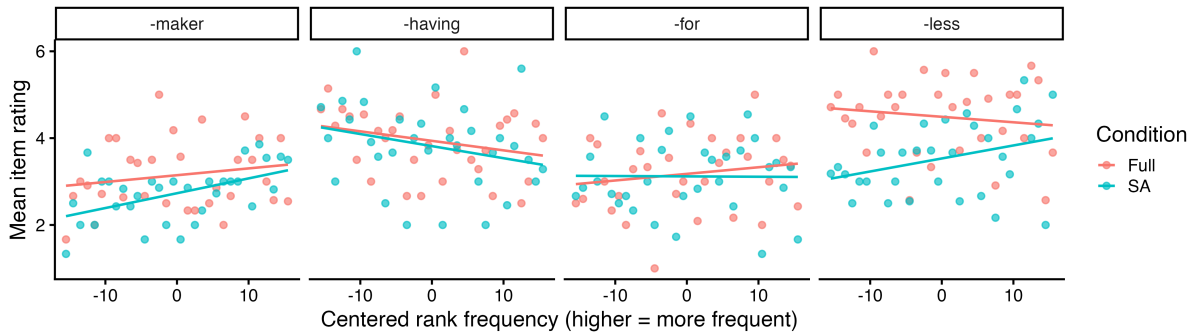


Figure 1: Observed mean item ratings by suffix and condition. Higher centered-rank values correspond to more frequent coordinations.

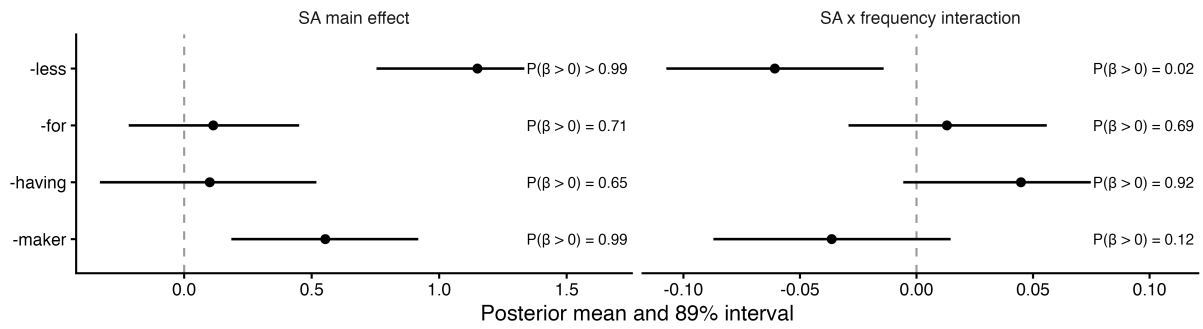


Figure 2: Posterior summaries from the suffix-specific Bayesian ordinal models. The left facet shows the SA main effect; the right facet shows the SA-by-frequency interaction. Labels at the right edge give $P(\beta > 0)$ for each coefficient.

interaction with frequency. In Figure 2, a positive sign for the SA effect suggests higher ratings for FULL relative to SA forms; a negative sign for the interaction effect suggests that frequency improves the ratings for SA forms more than it improves those for their FULL versions.

Results showed an overall decrease in the ratings for SA relative to FULL forms with '-less' ($\hat{\beta}$=1.15, 89%CrI [0.75, 1.53]) and '-maker' ($\hat{\beta}$=0.55 [0.18, 0.92]). There was no such a difference with '-having' or '-for'. The clearest evidence for an interaction between form and frequency was with '-less', evidenced by an increase in the acceptability of SA forms with higher frequency ($\hat{\beta}$=−0.06 [-0.11, -0.01]). There was no clear evidence for an interaction with '-maker' or '-for'. Lastly, '-having' showed a trend in the opposite direction, suggesting a decrease in the acceptability of SA forms with higher frequency; but this trend was not statistically reliable ($\hat{\beta}$=0.04 [-0.01, 0.10]).

Overall, there was clear evidence that the four suffixes tested here differ with respect to the overall SA penalty and how this penalty is modulated by frequency. This suggests that a simple narrative where frequency uniformly boosts acceptability is insufficient. Moreover, the cumulative-logit models we report above can only describe final ratings, which entangle structural parsing preferences with baseline response biases and inattention. To systematically reveal the latent variables contributing to the ratings and determine the exact role of frequency in syntactic choice, we use a Multinomial Processing Tree process model, which we discuss next.

3 Latent Modeling

3.1 Methods

It is widely accepted that experimental measures such as acceptability judgments or reading times are not transparent reflections of a single underlying process; rather, they are generated by a mixture of latent subprocesses (Ratcliff, 1978; Riefer and Batchelder, 1988; Batchelder and Riefer, 1999; Paape and Vasishth, 2022; Nicenboim et al., 2024). Consequently, inferences based only on condition means can underrepresent some cognitive processes while amplifying others. On this view, assuming a simple, one-to-one mapping between frequency and acceptability is too coarse: frequency may influence ratings indirectly by shifting structural choice, while the observed responses are simultaneously shaped by attention and response bias. The descriptive results above align with this point, because neither frequency nor the availability of derivational SA exerts a single linear effect across suffixes, suggesting a potential role for subprocesses in acceptance ratings.

To better account for several probabilistic latent subprocesses in the rating task described above, we fit a Bayesian Multinomial Processing Tree (MPT) model to the rating data. We make three main assumptions in the model. First, the structural choice between a base-generation and an ellipsis analysis is probabilistic rather than deterministic, and *probabilities for structural choices* can shift as a function of experience-based factors such as frequency (Jurafsky, 1996). The second assumption is that participants may not always be attentive in their responses. When they are inattentive, their responses are not random; instead, they are influenced by an a priori response yes/no tendency, which we model as a *subject-level response-bias*. Third, in addition to differences across participants, a single participant's attentional state may vary from trial to trial. While a participant may sometimes respond on the basis of the stimulus, they may sometimes fail to do so (Paape and Vasishth, 2022; Nicenboim et al., 2024). We model this as *trial-wise attentional variation*.

Before fitting the MPT, we converted 6-point ratings to binary accept/reject outcomes. The threshold was chosen empirically from filler items by fitting a Bayesian cumulative-logit model and evaluating each candidate threshold by its posterior balanced accuracy in separating grammatical from ungrammatical fillers. A threshold at 4 gave the best performance (balanced accuracy = 0.929, 90% interval: [0.914, 0.942]); ratings of 4–6 were treated as acceptance and 1–3 as rejection.

The MPT was then fit to these binary judgments. Let $a_{j[n]}$ denote subject j's attentiveness on trial n and $g_{j[n]}$ their filler-derived yes-bias. For SA items, the attentive state involves a further latent choice: with probability b_n the parser adopts a base-generation analysis and the SA item inherits full-form acceptance θ_n^{full}; with probability $1 - b_n$ it adopts an ellipsis analysis governed by θ_n^{ell}:

$$(9) \quad P(\text{yes}_n) = a_{j[n]} \left[b_n \, \theta_n^{\text{full}} + (1 - b_n) \, \theta_n^{\text{ell}} \right] + (1 - a_{j[n]}) \, g_{j[n]}$$

For FULL-form items $b_n = 1$ by assumption, which simplifies Equation (9) to (10):

$$(10) \quad P(\text{yes}_n) = a_{j[n]} \, \theta_n^{\text{full}} + (1 - a_{j[n]}) \, g_{j[n]}$$

The latent subprocesses of the MPT model shown in (7) and (8) are as follows.

Attention (a_j): Each participant's attentiveness probability, governing whether their response reflects the structural model or falls back to their default yes-bias, calculated by their responses to grammatical fillers.

Criterion-derived yes bias (g_j): Estimated from each participant's hits and false alarms on filler items via a signal-detection-theoretic criterion c_j, mapped to a yes-bias through $\Phi(-c_j)$. Using exclusively fillers ensures that response bias and attentiveness are identified independently of the critical SA items (Hautus et al., 2021).

Attentive acceptance (θ_n^{full}): Suffix-specific acceptance probability for FULL forms, with varying effects for subjects and items.

Structural route (b_n): For SA forms, the probability of adopting a base-generation analysis (Equa-

tion (9)). Under base generation, the SA item inherits θ_n^{full}; under ellipsis $(1 - b_n)$, acceptance is governed by a separate suffix-specific parameter θ_n^{ell}.

Frequency effect on b_n: The base-generation probability is modeled on the logit scale with a suffix-specific intercept and slope for centered rank frequency, plus varying effects for subjects and items. A positive slope means higher frequency favors base generation—the structural route that licenses derivational SA.

The MPT model was implemented directly in Stan (Carpenter et al., 2017).

3.2 Results

The MPT outperformed a maximal `brms` baseline on PSIS-LOO (ΔELPD = 7.47; Sivula et al. 2025; Vehtari et al. 2017), and is therefore used as the primary model. Nuisance parameters were well-identified: mean attentiveness was 0.81 ($SD = 0.01$), and the filler-derived criterion was mildly liberal ($M = -0.24$, $SD = 0.41$), corresponding to an inattentive yes-bias of 0.59 ($SD = 0.15$), ensuring that structural parameters do not absorb response bias.

Table 2 reports the structural parameters. SA-form acceptability (row 2) is not a free parameter: it is the weighted average $b \cdot \theta_{\text{full}} + (1 - b) \cdot \theta_{\text{ellip}}$, so the gap between rows 1 and 2 reflects both how penalized the ellipsis route is and how often it is taken.

Two patterns stand out. First, the SA penalty varies substantially across suffixes: largest for '-less' and '-maker', modest for '-having', and negligible for '-for'. The '-for' case shows why frequency cannot act as a direct acceptability booster: when both routes already yield poor acceptance, shifting route probability has little observable effect on ratings.

Second—and this is the central contribution of the MPT—frequency shifts $P(\text{base-gen})$ in the predicted direction for the two suffixes with the clearest SA penalty ($P(\beta{>}0) = 0.83$ for '-less'; 0.80 for '-maker'). The *-maker* effect had no surface signature in the ordinal model, because the low full-form acceptability (0.34) limits the observable gain from route-shifting. For '-having', the frequency effect runs negative ($P(\beta{>}0) = 0.12$), and for '-for', it is uninformative ($P(\beta{>}0) = 0.62$).

Overall, frequency is better understood as modulating structural route availability than as adding a uniform boost to surface acceptability.

4 General Discussion

This study investigated how lexical frequency modulates the acceptability of derivational suspended affixation and modeled the underlying mechanism using a Multinomial Processing Tree. The MPT's central contribution is to decompose what might look like a direct frequency-acceptability link into a frequency effect on structural route probability: higher-frequency items are more likely to receive a base-generation analysis, and acceptability then follows from that structural choice together with nuisance processes. Critically, this decomposition revealed a frequency effect on structural choice for '-maker' that was entirely absent from the surface ordinal ratings — demonstrating that condition means alone are insufficient for detecting latent structural preferences.

Not all suffixes showed the predicted frequency-driven increase in base-generation probability, and the MPT explains why. For '-for', base generation is already the modestly dominant parse at mean frequency ($P(b{>}0.5) = 0.69$), and both full-form and SA-form acceptability are low—a configuration in which frequency-driven route-shifting yields little observable leverage on ratings. For '-having', frequency moved base-generation probability in the negative direction ($P(\beta{>}0) = 0.12$), possibly reflecting this suffix's relatively higher ellipsis-route acceptability compared to the other suffixes. These patterns are not unexplained residuals: they follow naturally from the model, because the observable consequence of route-shifting depends on the suffix-specific acceptability.

A deeper question raised by these results is why certain suffixes resist an ellipsis analysis in the first place. Theoretical frameworks such as Distributed Morphology (Halle and Marantz, 1993, 1994) have argued that the traditional derivational/inflectional divide is better conceptualized as a continuum determined by syntactic attachment height and the position of functional heads in the syntactic spine. Suffixes that attach low in the structure — close to the root — are more lexically integrated and therefore a poorer target for post-syntactic deletion. This predicts that suffixes like '-less' and '-maker', whose base-generation probability is most strongly modulated by frequency, are precisely those whose attachment height makes ellipsis less viable. Making this connection precise by independently characterizing the attachment heights of Turkish derivational suffixes is an important direction for future

Parameter	Suffix	M [Lower, Upper]	Interpretation
Full-form acceptability	*-sIz* '-less'	0.86 [0.79, 0.92]	Attentive-state $P(\text{accept})$ for full forms
	-cI '-maker'	0.34 [0.24, 0.44]	
	-lI '-having'	0.77 [0.69, 0.85]	
	-lIk '-for'	0.36 [0.26, 0.46]	
SA-form acceptability	*-sIz*	0.49 [0.37, 0.62]	Marginal $P(\text{accept})$ at mean frequency
	-cI	0.19 [0.11, 0.29]	
	-lI	0.65 [0.53, 0.76]	
	-lIk	0.32 [0.23, 0.42]	
Ellipsis acceptability	*-sIz*	0.33 [0.17, 0.52]	Attentive-state $P(\text{accept})$ via ellipsis route
	-cI	0.11 [0.04, 0.21]	
	-lI	0.52 [0.32, 0.70]	
	-lIk	0.24 [0.11, 0.38]	
$P(\text{base-gen})$ at mean freq.	*-sIz* ($P(\beta{>}0.5){=}0.08$)	0.30 [0.11, 0.54]	Ellipsis strongly dominates
	-cI ($P(\beta{>}0.5){=}0.26$)	0.38 [0.12, 0.71]	Ellipsis dominates
	-lI ($P(\beta{>}0.5){=}0.50$)	0.50 [0.19, 0.81]	Completely uncertain
	-lIk ($P(\beta{>}0.5){=}0.69$)	0.60 [0.23, 0.89]	Base-gen. modestly dominant
Effect of freq. on $P(\text{base-gen.})$	*-sIz* ($P(\beta{>}0){=}0.83$)	$+0.08$ [-0.06, $+0.26$]	suggestive positive effect
	-cI ($P(\beta{>}0){=}0.80$)	$+0.15$ [-0.16, $+0.45$]	suggestive positive effect
	-lI ($P(\beta{>}0){=}0.12$)	-0.20 [-0.49, $+0.08$]	suggestive negative effect
	-lIk ($P(\beta{>}0){=}0.62$)	$+0.06$ [-0.33, $+0.43$]	uncertain

Table 2: Posterior means and 89% credible intervals for key MPT parameters. In the lower section, parenthetical values next to each suffix are $P(\beta{>}\text{chance})$: the posterior probability that base generation is the dominant parse at mean frequency. In the frequency effect section, parenthetical values are $P(\beta{>}0)$: the posterior probability that the effect of frequency on base-generation probability is positive. The frequency effect is on the logit scale.

work.

We identify two further directions for testing and extending the account. First, the base generation and ellipsis parses make different scopal predictions: under base generation, the derivational suffix takes scope over the full coordination, applying to both conjuncts jointly, while under ellipsis each conjunct receives the suffix independently. These parses are therefore distinguishable by their interpretation. For example, *kitap ve dergi-siz* under a base-generation parse means 'lacking books-and-magazines (jointly)'; under ellipsis it means 'lacking books and lacking magazines (separately)'. If frequency biases readers toward base generation as the MPT suggests, higher-frequency items should preferentially support the joint, wide-scope reading in a forced-choice interpretation task. Such a task would provide direct, independent evidence that frequency operates on parse selection rather than on acceptability directly—the core theoretical claim of this paper.

Second, the present account predicts that the frequency effect on base-generation probability should only arise for affixes that are incompatible with the ellipsis route. Our discussion of what blocks ellipsis has remained informal; making it precise is both theoretically necessary and empirically tractable. Ellipsis analyses of SA have been discussed extensively in the syntactic literature (e.g., Despić, 2017; Erschler, 2018; Guseva and Weisser, 2018; Türk, 2025), and connecting the present computational results to those structural debates would sharpen the predictions considerably.

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